Lecture 20: AR(p) Models and Identification of AR Models

IEOR E4709: Statistic Analysis and Time Series
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1 AR(p) Model

The auto-regressive model of order p, denoted as $AR(p)$, is

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t, \tag{1}$$

where $\{a_t\}$ is a white noise sequence, i.e., an i.i.d. sequence with mean zero and variance σ_a^2. The quantities ϕ_i's are constants. Moreover, it is assumed that $\Phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$, $z \in \mathbb{C}$, where $\mathbb{C}$ denotes the set of complex numbers, has no zero within the unit circle, i.e. $\Phi(z) \neq 0$ for any z such that $|z| \leq 1$. This condition is necessary to ensure that $\{r_t\}$ is weakly stationary.

The conditional mean of r_t is given by

$$\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right] = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i}.$$

The unconditional mean is given by

$$\mu := \mathbb{E}\left[\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \mathbb{E}\left[r_{t-i}\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \mu. \tag{2}$$

This leads to $\mu = \frac{\phi_0}{1-\sum_{i=1}^{p}\phi_i} = \frac{\phi_0}{\Phi(1)}$. Using this equation to replace ϕ_0 in Eq. (1), we can rewrite the $AR(p)$ model as

$$r_t - \mu = \sum_{i=1}^{p} \phi_i (r_{t-i} - \mu) + a_t.$$

For each $l \geq 0$, we have

$$\begin{aligned} \gamma_l &= cov(r_t, r_{t-l}) = cov(a_t, r_{t-l}) + \sum_{i=1}^{p} \phi_i cov(r_{t-i}, r_{t-l}) \\ &= cov(a_t, r_{t-l}) + \sum_{i=1}^{p} \phi_i \gamma_{l-i}. \end{aligned}$$

When $l = 0$, we have that $cov(a_t, r_{t-l}) = cov(a_t, r_t) = \sigma_a^2$, and consequently

$$\gamma_0 = \sigma_a^2 + \sum_{i=1}^{p} \phi_i \gamma_{-i} = \sigma_a^2 + \sum_{i=1}^{p} \phi_i \gamma_i = \sigma_a^2 + \gamma_0 \sum_{i=1}^{p} \phi_i \rho_i,$$

where we have used that the lag l autocorrelation is $\rho_l = \frac{\gamma_l}{\gamma_0}$. Therefore, the unconditional variance is given by

$$\gamma_0 = \frac{\sigma_a^2}{1 - \sum_{i=1}^{p} \phi_i \rho_i}. \tag{3}$$

For $l \geq 1$, we have that $cov(a_t, r_{t-l}) = 0$, and consequently

$$\gamma_l = \sum_{i=1}^{p} \phi_i \gamma_{l-i} = \sum_{i=1}^{p} \phi_i \gamma_{|l-i|}.$$

Therefore, we obtain

$$\rho_l = \sum_{i=1}^{p} \phi_i \rho_{|l-i|}$$

The ρ_l's, $l = 1, \ldots, p$, satisfy the **Yule-Walker equation:**

$$\underbrace{\begin{pmatrix} 1 & \rho_1 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \cdots & \rho_{p-2} \\ \vdots & \vdots & \vdots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \cdots & 1 \end{pmatrix}}_{correlation\ matrix\ of\ (r_1, r_2, \ldots, r_p)} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \cdots \\ \phi_p \end{pmatrix} = \underbrace{\begin{pmatrix} \rho_1 \\ \rho_2 \\ \cdots \\ \rho_p \end{pmatrix}}_{ACF} \tag{4}$$

The estimation of the $AR(p)$ model proceeds similarly to the estimation of the AR(1) model. Suppose r_t, $t = 1, \ldots, T$ are observed. We can use the method of moments as follows:

- Use the Yule-Walker equation and the sample correlation to estimate ϕ_i, $i = 1, \ldots, p$.
- We use Eq. (2) for the unconditional mean and the sample mean estimate for μ, to estimate ϕ_0.
- We use the sample variance to estimate γ_0. We then use Eq. (3) along with the sample correlations and the coefficients ϕ_i estimated in the previous step to estimate σ_a^2.

Alternatively, assuming that r_j, $j = 0, \ldots, t-1$ are such that r_t conditional on $\mathcal{F}_{t-1}$ (where we recall that $\mathcal{F}_{t-1}$ is the filtration generated by the returns $r_1, r_2, \ldots, r_{t-1}$) is normally distributed with mean $\phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i}$ and variance σ_a^2, we can use the MLE method. The objective is to maximize the conditional likelihood of r_t, $t = 1, \ldots, T$, given $r_0, r_{-1}, r_{-2}, \ldots, r_{-(p-1)}$.

2 Identification of AR Models

How do we specify the order of an AR model? One approach is to use the partial autocorrelation function (PACF). What is the PACF? The PACF at lag k is defined as the correlation between r_t and r_{t-k} after removing the effect of the variables in between: $r_{t-1}, \ldots, r_{t-k+1}$.

For example, the formula for the partial correlation between r_t and r_{t+2}, i.e., the correlation of r_t and r_{t+2} conditional on r_{t+1} is given by

$$\rho_{r_t,r_{t+2}|r_{t+1}} = \frac{\rho_{r_t,r_{t+2}} - \rho_{r_t,r_{t+1}}\rho_{r_{t+1},r_{t+2}}}{\sqrt{1-\rho^2_{r_t,r_{t+1}}}\sqrt{1-\rho^2_{r_{t+2},r_{t+1}}}}.$$

For an AR(1) model, $\rho_{r_t,r_{t+2}} = \phi_1^2$ and $\rho_{r_t,r_{t+1}} = \rho_{r_{t+1},r_{t+2}} = \phi_1$. Hence, $\rho_{r_t,r_{t+2}|r_{t+1}} = 0$.

Remark 2.1. *Observe that the PACF at lag 1 is the same as the lag-1 autocorrelation function. However, for lags $k > 1$, the PACF at lag k and the ACF at lag k can be quite different.*

PACF is at the heart of the order identification procedure for an AR model. Consider a sequence of AR models:

$$\begin{aligned}
r_t &= \phi_{0,1} + \phi_{1,1}r_{t-1} + e_{1t} \\
r_t &= \phi_{0,2} + \phi_{1,2}r_{t-1} + \phi_{2,2}r_{t-2} + e_{2t} \\
r_t &= \phi_{0,3} + \phi_{1,3}r_{t-1} + \phi_{2,3}r_{t-2} + \phi_{3,3}r_{t-3} + e_{3t} \\
r_t &= \phi_{0,4} + \phi_{1,4}r_{t-1} + \phi_{2,4}r_{t-2} + \phi_{3,4}r_{t-3} + \phi_{4,4}r_{t-4} + e_{4t},
\end{aligned}$$

where $\phi_{0,j}, \phi_{i,j}$, and $\{e_{jt}\}$ are, respectively, the constant term, the coefficient of r_{t-i}, and the error term of an $AR(j)$ model. These models are in the form of a multiple linear regression and can be estimated using the least squares method. Hence, we can use the idea of F tests in multiple linear regression analysis.

The estimate $\hat{\phi}_{1,1}$ of the first equation is called the lag-1 sample PACF of r_t. The estimate $\hat{\phi}_{2,2}$ of the second equation is the lag-2 sample PACF of r_t. The estimate $\hat{\phi}_{3,3}$ is the lag-3 sample PACF of r_t and so on. From the definition, we see that the lag-2 PACF $\hat{\phi}_{2,2}$ is the added contribution of r_{t-2} to r_t over the AR(1) model $r_t = \phi_0 + \phi_1 r_{t-1} + e_{1t}$. The lag-3 PACF is the added contribution of r_{t-3} to r_t over an AR(2) model, and so on. Hence, for an $AR(p)$ model, the lag-p sample PACF should not be zero, but $\hat{\phi}_{j,j}$ should be close to zero for all $j > p$.

For a stationary Gaussian $AR(p)$ model, we can show that the sample PACF ha the property that $\hat{\phi}_{p,p}$ converges to ϕ_p, and $\hat{\phi}_{l,l}$ converges to 0 for all $l > p$, as the sample size T goes to infinity. Moreover, the asymptotic variance of $\hat{\phi}_{l,l}$ is $1/T$ for $l > p$. All these results state that, for an $AR(p)$ series, the sample PACF *cuts off* at lag p.